University of Florida, Department of Mathematics

Topology PhD Exam, May 2012

Work the following problems and show all work. Support all statements to the best of your ability.

1. Does there exist a compact space that contains a discrete infinite subspace?

2. Let X and Y be connected spaces and X, Y is not the one-point space. Show that $X \times Y \setminus \{z\}$ is connected for any $z \in X \times Y$.

3. Prove that $(0, 1)$ and $[0, 1)$ are not homeomorphic.

4. Show that the 2-sphere S^2 is not a retract of the projective plane, as well as the projective plane is not a retract of S^2.

5. Let $GL(n)$ be the space of all invertible real $n \times n$-matrices. IS GL_n is compact? connected?

Answer the following with complete definitions or statements or short proofs.

6. Explain why every map $S^n \to S^1$, $n > 1$ is null-homotopic.

7. State the Lefschetz Fixed Point Theorem. Explain that every continuous map $D^n \to D^n$ has a fixed point.

8. State the Mayer–Vietoris sequence for homology.

9. Show that the spaces $S^2 \times S^2$ and $S^2 \vee S^2 \vee S^4$ are not homotopy equivalent.

10. Does every function $f : \mathbb{N} \to \mathbb{R}$ admits a continuous extension $\bar{f} : \beta\mathbb{N} \to \mathbb{R}$ of the Stone–Čech compactification?

11. Formulate the Invariance of Domain Theorem.

12. Let M be a closed 4-dimensional simply connected manifold. What can you say about $H_3(M)$?

13. Does there exist a cover space of the figure eight that has the non-trivial abelian fundamental group?

14. Let X be a compact Hausdorff topological space. Describe all compact subsets of the space $C(X, \mathbb{R})$ of continuous functions $X \to \mathbb{R}$ in uniform topology.

15. State the Universal Coefficient Theorem for Cohomology.